

History of Computing in India: 1955–2010

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The history of computing in India is inextricably intertwined with two interacting forces: the political climate and government policies, mainly driven by the technocrats and bureaucrats who acted within the boundaries drawn by the political party in power. Four break points occurred in 1970, 1978, 1991, and 1998 that changed the direction of the development of computers and their applications and affected the growth of IT in India.

The history of computing in India began in 1955, when a HEC-2M designed by A.D. Booth was installed at the Indian Statistical Institute (ISI) in Calcutta.¹ In 1955, a team headed by Rangaswami Narasimhan started designing and fabricating a digital computer at the Tata Institute of Fundamental Research (TIFR) in Bombay.² At this time, only a small number of scientists and engineers in India knew about computers. In 2010, the year we end our history, there were more than 2.4 million people employed in computer-related jobs and more than 60 million PCs were in use. Information technology (IT) contributed 6.4 percent of the gross domestic product (GDP) of India, and IT services became the fastest growing segment among export industries. IT grew by 22.7 percent in 2010 with aggregate export revenue of US\$50 billion and domestic revenue of US\$17 billion.³ Undoubtedly, this was an exciting journey from 1955 to 2010, although not smooth and steady. In the 1960s and 1970s, there was a lot of trepidation about the use of computers and their impact on employment. Questions were asked about whether computers were relevant for an overpopulated, poor country. From this uncertain beginning, India reached a stage in 1998 when the then prime minister of India declared IT as “India’s tomorrow.” How did India reach this state? What were the contributing factors? What lessons did India learn?

The development of computing in India is inextricably intertwined with two interacting

forces: the political climate and the government policies, mainly driven by the technocrats and bureaucrats guided by the political party in power and some external forces. There were four “break points” in the development of computers and their applications in India as a result of significant events that occurred in 1970, 1978, 1991, and 1998. This article explains why these breaks occurred and how they affected the growth of IT in India.

Several studies on various aspects of the development of computers in India are available in the literature. Dinesh C. Sharma’s book, *The Long Revolution—The Birth and Growth of India’s IT Industry* gives a detailed account of the history of IT in India from a journalist’s perspective.⁴ In an earlier book, C.R. Subramanian describes computer technology in India before 1990 and highlights the weaknesses inherent in the government policy of planned development of computers.⁵ The book has detailed statistics and excerpts from government archives. Ramesh Subramanian traces the history of IT in India by talking to five IT professionals representing different groups—the government, education, research, and industry.⁶ Joseph M. Grieco analyzes how India negotiated with the international computer industry to preserve its national interest without becoming subservient to multinationals.⁷ Balaji Parthasarathy discusses how India’s domestic policy initiatives enabled the Indian software

industry to grow rapidly.⁸ Ross Basset analyzes the impact of computer education initiatives taken by IIT, Kanpur, and other IITs and how this mutually helped India and the United States.⁹ Peter Walcott and Seymour Goodman document the growth of computer networks in India from 1980 to 2003.¹⁰ In an earlier article, I give a brief retrospective of IT in India during the period 1965–1999.¹¹ Articles in the first part of the book, *Homi Bhabha and the Computer Revolution*, edited by Rudrapatna K. Shyamasundar and M. Ananth Pai, present a historical perspective on the development of computer technology in India.¹² The book also includes articles written by pioneers of the Indian computer and communication industry and provides a wealth of first-hand material. James W. Cortada analyzed the international diffusion of IT from 1940 to late 1990s.¹³ His book includes a chapter on IT diffusion in India from the perspective of a western professional historian.

To the best of my knowledge, no article has been published on the history of IT in India, identifying break points, why they occurred, and how they changed the pattern of growth. This article aims to cover this gap. It summarizes a monograph I wrote that was published on the Web by the IEEE Computer Society.¹⁴ I wrote this article not as a professional historian but as one who participated in the development of IT in India from the 1960s to date as an academic, an IT consultant to a large number of industries, and a member of numerous committees of the Government of India involved in policymaking. Hopefully this article will kindle the interest of professional historians to undertake a deeper study of India's IT history.

Some Facts about India

India is the second most populous country in the world, with a population of 1.2 billion. It is an old civilization but a young country, with a median age of 26.2 years. India has adopted a mixed economy—a number of public sector companies control oil, natural gas, and heavy industries. Currently there is a thriving private sector. IT is dominated by the private sector. Privatization of major industries began in the early 1990s. Economic growth rate was slow (around 3.8 percent of the GDP) for the first 30 years after independence in 1947, but it was above 7 percent between 2002 and 2010. India's current GDP (based on purchasing power parity) is US\$4 trillion, the fifth highest in the world,¹⁵

although the GDP per capita is only US\$3,500. The country faces long-term challenges—inadequate physical and social infrastructure, widespread poverty, wide disparity in the quality of education offered, limited rural employment opportunities, and waste in government spending.

Communication facilities in India have improved rapidly since the mid-1980s. Currently, India has more than 40 million landlines and 850 million mobile phone subscribers. India builds and launches its own satellites. Unlike the deep penetration of mobile phones, however, that of the Internet is limited. Although the number of Internet users is estimated to be around 100 million (the third largest in the world), the Internet reaches only about 8.4 percent of the population.

A major problem faced by India is an endemic shortage of electricity. Electric power generation is primarily controlled by the states, and inadequate investment in this important sector has adversely affected all industries, including IT.

Manufacturing industry such as automobiles has matured owing to collaboration with Japanese and western companies. Most electrical consumer goods are manufactured locally and their quality is good. Although local manufacturing of PCs and sophisticated ICs is almost nonexistent, there is a thriving industry designing ICs and other sophisticated electronic hardware for western customers. The world class software services industry is quite mature.

Although Hindi is the official language, English, which was introduced by the British, continues to be the language used by the government and the judiciary. The medium of instruction in science and engineering courses in colleges is also English. It is estimated that 125 million people in India have a working knowledge of English.¹⁶ The large number of English-speaking persons in the 20–50 years age group (estimated to be more than 50 million) has led to a number of IT-based services from western countries being outsourced to India.

The dream of Jawaharlal Nehru, the first Prime Minister of India, was to make India a social democracy with a mixed economy—that is, he envisioned the coexistence of private and state enterprises. His government believed in planned development and constituted the Planning Commission. The primary function of this commission was and continues to be the drafting of five-year plans for

the growth of various sectors of the economy and to allocate resources (see www.planning-commission.gov.in/aboutus/history). Jawaharlal Nehru had great faith in science and technology as engines of economic growth. He was convinced that India required rapid industrialization to reduce the abysmal poverty of its people. India's Parliament passed the Scientific Policy Resolution in 1958, the full text of which may be found at www.dst.gov.in/stsysindia/spr1958.htm.

Another problem India faced was a balance of payments deficit. Setting up heavy industries demanded imports, and most of the required petroleum products had to be imported. Exports comprised mostly raw materials such as cotton and minerals. There was dearth of foreign exchange (that is, hard currency such as US dollars earned through export), which dictated many of the government's policies. Any private industry requiring import using scarce foreign exchange was subjected to close scrutiny. A company was required to earn foreign exchange through export in order to import goods. After liberalization in 1991, this situation changed and imports of capital goods including computers became easier, and Indian companies are now allowed to invest abroad. However, the rupee is not a freely convertible currency.

Laying the Foundation (1955–1970)

The first group to build a digital computer in India was led by R. Narasimhan at the TIFR, Bombay. This group started building a computer, the TIFR Automatic Computer (TIFRAC), in 1955 and completed it in 1959. It had 2,048 word core memory (40-bit word, 15 microsecond cycle time), an arithmetic unit with some innovations,^{2,12} a paper tape I/O, and a cathode ray tube output unit. An assembler was written for it. It was used for solving physics problems in-house. It was also used by the scientists of the Atomic Energy Establishment and some universities. This project proved that Indians could design a computer and use it effectively. In 1959 there were no commercial digital computers in India.

The first computer with a Fortran compiler to be installed in an educational institution in India was an IBM 1620 at the Indian Institute of Technology Kanpur (IITK) in August 1963. It was used to spread computer education in India. Details of the contribution of IITK to the development of IT in India may be found elsewhere.^{9,17} The first high-performance computer installed in India was a CDC 3600-160A at the TIFR in mid-1964^{18,19}

and was used extensively by several universities and research institutes.

During the 1960s, computers were not considered a priority area that deserved foreign exchange outflow. IBM and the British Tabulating Machines (which was later named International Computers Ltd., ICL) were already selling mechanical accounting machines in India. Because importing computers using foreign exchange was difficult, IBM and ICL applied for licenses to manufacture computers in India. IBM started manufacturing punch card machines and exported them. With the foreign exchange earned, it imported used 1401 computers, refurbished them, and rented them to organizations in India. By 1970, the IBM 1401 was the most popular computer in India, with 80 installations. (In 1971, the IBM 1401 was phased out in the United States.) The annual rental charge was higher than it was in the US. In its defense, IBM asserted that it was selling a computing service rather than just renting computers. And to its credit, IBM's service was excellent, and it recruited and trained good technical and sales persons from India. IBM trained and nurtured a whole generation of maintenance engineers and programmers.⁵

Self-Reliant Growth of the Computer Industry (1970–1977)

During the 1950s, electronics was not considered an important industry by the Government of India and there were no specific policy initiatives. After the border skirmish with China in 1962, in which India lost some territory, it was realized that modern electronics and communication equipment were essential in defense preparedness and for the long-range industrial growth of India. The Government of India constituted a committee in 1963 with Homi Bhabha as its chairman to examine the area of electronics in depth and prepare a plan for its development. One of the main recommendations of the committee was to establish an electronics commission (EC) with wide financial and executive powers to make quick decisions to promote electronics and computer industry²⁰ and the Department of Electronics (DoE) to execute the policies formulated by the EC.

Political Environment and Government Policies

The Congress party, which spearheaded the independence movement, governed India uninterrupted from 1947 to 1977. The public sector had a preeminent role in the economy. All industries were centrally controlled. India

had a perpetual balance of trade deficit. The rupee was not convertible. Thus, obtaining hard currency for importing equipment required elaborate justification.

There was also some trepidation regarding the impact of computers on employment. A committee chaired by Member of Parliament Vinayak M. Dandekar examined the issue in 1972 and prescribed strict controls on introducing computers in industry and government departments.²¹ Parliament's Public Accounts Committee in its 1975 report was also cautious.²² The committee opined that the use of computers could lead to efficiency, higher profits, and faster economic growth, but only in the long run. The committee felt that in India, with its large-scale unemployment, the use of computers and other sophisticated machines for labor-saving applications might not be desirable and might even be detrimental. It recommended that the government take into account the social cost of computerization and evolve a principled and positive approach on computerization, keeping in view the overall national interest.

The Department of Electronics was constrained by these observations and had to move cautiously. In the 1970s, the DoE drafted an elaborate set of rules for importing computers, which led to a delay of one to three years in obtaining clearance to import.²³ In a fast-changing field such as computers, this delay was unacceptable. This restriction particularly hurt companies that wanted to import computers for software export. It also adversely affected the projects of many industries, scientific research laboratories, and universities.

The other major political event that affected the development of computers was the war with Pakistan in 1971, which ended with the creation of Bangladesh. Richard Nixon, the US president, favored Pakistan, and this resulted in embargos on electronics and computer imports from the United States. The first nuclear test by India in 1974 further aggravated Indo-US relations, leading to an embargo on the import of electronic equipment using advanced technology, which included high-end computers and sophisticated software used in science and engineering.

Bhabha Committee's report of 1968 had recommended that computers, other than the larger ones, must be locally manufactured. It opined that "attaining self-sufficiency in systems engineering and fabrication is of

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fundamental importance from the point of view of the defense and security of our country." It also suggested that, from a long-range perspective, India should upgrade its capability to design and manufacture smaller computers. Consequently, the DoE fully funded a computer division in the public sector company Electronics Corporation of India Limited (ECIL) to manufacture and market computers.²⁴ Meanwhile, computer technology was rapidly progressing in the west. With the development of large-scale integration (LSI), the price of computers was falling. The DoE was expected to announce a "minicomputer policy" that would formulate the ground rules for private companies to manufacture computers; however, it delayed the announcement because it was concerned about the requirement of hard currency for importing components and peripherals. The DoE was also protecting ECIL, which it had funded.

Government Initiatives

ECIL designed TDC 12, a 12-bit real-time minicomputer, in 1969. Meanwhile, technology was changing. TDC 12 was upgraded in 1974 to TDC 312 and was followed by TDC 316, a 16-bit computer built in 1975. In 1978, ECIL manufactured a microprocessor-based system MICRO 78. ECIL sold 98 computers between 1971 and 1978, mostly to government laboratories and universities.²⁵ An important contribution of ECIL was providing ruggedized computers for the Indian Air Force's Air Defense Ground Environment Systems (ADGES). Because of their sensitive nature, these systems had to be designed and fabricated with Indian know-how and in

institutions located in India. Import would have been difficult because sensitive information had to be revealed to vendors, and India was often subjected to sudden embargos on electronic systems by the US and its allies. These systems were deployed along India's borders to detect intrusion by unfriendly aircraft. Each system used three TDC 316 computers, which were ruggedized with components adhering to MIL specifications. The systems with radars developed by the Defence Electronics and Radar Development Establishment in Bangalore were designed by a team at TIFR, Bombay, and 25 of these were deployed by the Indian Air Force from 1969–1984. The published literature does not mention this important contribution by ECIL (see www.tifr.res.in/~sanyal/national.html). The system could have been modified for use in air traffic control systems, but ECIL did not pursue this opportunity. (A brief history of ECIL appears elsewhere.²⁶)

The DoE was established with the primary objective of promoting the development of electronics and the computer industry. However, between 1970 and 1977, the general perception was that it was playing more of a regulatory role and protecting ECIL. The political environment^{21,22} and the scarcity of hard currency forced the DoE to take certain actions, particularly in regulating the import of computers and allowing manufacture by the private sector.

Regardless of this perception, the DoE did play a promotional role in a number of areas. Important initiatives taken by the DoE during this period include establishing the National Center for Software Development and Computing Techniques in 1972, establishing the National Informatics Center in 1975 to assist in the government's e-governance initiatives, funding the Army Radio Engineering Network, establishing regional computing centers with large imported mainframes in some universities, funding the ADGES project described earlier, and establishing the Computer Maintenance Corporation (CMC) to maintain imported computers,¹⁴ including IBM computers left behind by IBM.

Private Sector Enters the Computer Industry (1978–1990)

In 1973 India had enacted the Foreign Exchange Regulation Act (FERA) to conserve foreign exchange. Under this act, foreign companies, except those considered essential, were to dilute equity to 40 percent and take an Indian partner. IBM was asked to fall

in line. It refused and left India. (A detailed discussion of IBM's negotiations with the Government of India is available elsewhere.²⁷) IBM's decision to leave India and the announcement of a long-pending mini-computer policy opened up the industry for the entry of private entrepreneurs and the end of ECIL's monopoly.

Even though the political environment was choppy during this period, Rajiv Gandhi, who belonged to the post-independence generation, was open minded regarding private enterprise and ushered in significant changes when he became prime minister.

Political Environment and Government Policies

The period from 1978 to 1990 was one of political instability in India. After two unstable coalition governments, Indira Gandhi returned as prime minister in 1981. In 1984, Indira Gandhi was assassinated and her son Rajiv Gandhi took over as prime minister. He won the next general election and was prime minister from 1984 to 1989. In the general election of 1989, the Congress Party lost and a coalition government returned. This coalition was unstable and fell in 1990.

In spite of the political instability, computing progressed at a much faster rate during 1978–1990 than during 1970–1978. The period started with the appointment of a committee, chaired by Mantosh Sondhi,²⁸ whose task was to review the progress of electronics and computing. The Sondhi committee suggested permitting private manufacturers to manufacture computers. It opined that the major emphasis in the development of the minicomputer/microprocessor industry would be on setting up systems engineering companies, which were not necessarily engaged in the manufacture of central processing units or peripherals.

Two other events had a far-reaching impact on the government policy. At Rajiv Gandhi's insistence, all the clerical chores of the 1982 Asian Games held in Delhi were computerized using locally made computers and software developed by NIC. It was a resounding success and convinced Rajiv Gandhi of the importance of computers. A liberalized policy on minicomputers was announced in 1984 as soon as Rajiv Gandhi became prime minister. It allowed private sector companies to manufacture 32-bit machines, removed constraints on the number of computers a company could manufacture, allowed assembled boards with microprocessors and interface electronics to

be imported along with application software, and reduced the import duty.²⁹ The procedures for importing mainframes were simplified. The policy of allowing import of fully assembled boards changed manufacturing to assembly and system integration.

Software exports were promoted by recognizing software development and services as an industry. This recognition led to many fiscal and tax concessions. The sending of engineers abroad to develop and maintain software for clients at their sites and the profits earned thereby were recognized as "software exports." In 1986, a more liberalized software policy was announced that gave further incentives for software export.³⁰ Computers and software tools used to develop software for export could be imported duty free. Software developed in India could be exported using communication systems such as satellite and cable.

Government Initiatives

The period from 1978 to 1990 was also marked by a number of initiatives taken not only by the DoE but also by other organizations that accelerated the use of computers. These initiatives included starting undergraduate courses in computer science at IITs; starting a new degree course, the master of computer applications (MCA), to educate college graduates in systems analysis and design; funding the establishment of computer-aided design and computer-assisted management centers at elite educational institutions; funding centers to develop knowledge-based computer systems; starting the Center for the Development of Advanced Computing to develop high-performance parallel computers; allowing the use of satellite communication for software development from India by multinationals; and promoting the use of computers by banks and other financial institutions.

In addition, the initiation of two projects requires special mention. One was the project to computerize the Indian Railways seat/berth reservation system, which began in 1984 and was completed in 1986. India has one of the largest railway networks in the world. In 1984, Indian Railways handled more than 5 million passengers travelling in more than 600 long-distance trains with around 50,000 reservation requests. Passengers had to stand in long queues to obtain reservations. CMC developed a reservation system and implemented it at the New Delhi booking office. It had 50 counters, and customers could go to any counter to get a

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reservation for any train.³¹ This system was a huge success and was highly appreciated by the general public because it saved them enormous time. The entire software effort was by local programmers with no "foreign consultants." The reservation system using computers was an eye opener to the general public because it demonstrated the advantages of using computers. There was an attitudinal change among both the general public and white collar workers about computerization. This was the beginning of the acceptance of computers and the realization that, in a country with large volumes of data to be processed, the use of computers is inevitable. Some bank unions also accepted computerization that they had opposed earlier.

The second major initiative taken by the DoE was the establishment of software technology parks. STPs provided infrastructure such as buildings, work stations, continuous uninterrupted power supply, and satellite communication links to software companies located in these parks. Software companies were able to develop software on the computers of their overseas customers using the satellite communication links. Because the investment required to set up a company in an STP was low, this initiative allowed many small entrepreneurs to enter the software services export business. The first STP was established in Bangalore in 1990. STPs were set up later in many other cities and incorporated as STPI (www.stpi.in), controlled by the DoE.

Consequences of Government Policy

The minicomputer policy of 1978 opened up the computer industry and saw the

emergence of a number of technical entrepreneurs who started computer manufacturing operations. Among the most notable were the DCM, HCL, WIPRO, PSI, and ORG systems. Unix was the operating system of choice because it was easily available and was standardized for the computers to be installed in banks (a large market segment) by a committee headed by Chakravarti Rangarajan, who was the deputy governor of the Reserve Bank of India. Thus during 1978–1984 a large number of systems programmers in India became experts in Unix development and use. They innovated and adapted Unix to work on computers with small memory. They also adopted C widely because Unix was written in C. This expertise came in handy when India entered the software services export market in the late 1980s.

The 1984 liberalization policy had a deeper impact. Within two years of the announcement of the new policy, the growth of computers went up by 100 percent, and the cost went down by 50 percent.⁵ The growth was due to the liberal import of populated boards. Manufacturers had to only do systems engineering and develop appropriate software. This made PCs affordable and spread the use of computers. Another consequence of the liberalization of 1986 was the entry of many multinational companies that collaborated with the local companies as minority partners.

The liberalization of the import of computers and software enunciated in the 1986 policy gave an impetus to the software export industry. Export earnings were insignificant in 1978, but they increased to US\$128 million in 1990. Establishing STPs by the DoE allowed many entrepreneurs to enter the software export business. Permitting Texas Instruments (TI) to open an offshore software development center in Bangalore and allowing TI to link with its Dallas center via satellite communication set a new trend. GE computer services, the Citibank software group, American Express, Cadence, and many others set up software development centers in India to take advantage of lower costs and the high quality of software developers. This period also saw the establishment of two industry groups: the Manufacturer's Association for IT (MAIT) in 1982, and the National Association of Software and Services Companies (NASSCOM) in 1988. A detailed account of the contributions of NASSCOM to the growth of India's software industry is available elsewhere.³²

Liberalization of the Economy and Software Export Growth (1991–1997)

This period was an exciting one for the IT industry in India. India started with a near default economic situation in 1991 that forced the government to open the economy and remove many of the controls on industry. The rupee was devalued (from Rs.17.5 to Rs.26 per US dollar) and foreign investments and industries were welcomed. This, along with the fortuitous circumstances of the need to fix the Y2K bug, the Euro conversion requirement, and a technically savvy influential Indian diaspora in the US opened opportunities to Indian software companies. This, coupled with the advent of fast satellite communication, the availability of human resources with strong English-language skills, quality consciousness of software companies, and project management expertise allowed the industry to get remunerative software services contracts from the west, particularly the US. Export earnings increased from US\$128 million in 1990 to US\$1,759 million in 1997.³³ The average annual growth rate of 45 percent of the software services export during this period was spectacular. The industry also provided employment to 160,000 software engineers. Thus, by 1997 there was a lot of optimism about the future of IT in India.

Political Environment and Government Policies

In the 1991 general election, the Congress Party returned to power. The new government faced a difficult economic situation. India was about to default payment of its external debts. The country was bailed out by the International Monetary Fund and the World Bank, which imposed a set of conditions that forced India to liberalize its economy. That liberalization changed the course of the history of computing in India. In the 1996 general election, the Congress Party was defeated. A coalition was formed, but did not last, and there was political instability with another coalition coming to power. Fortunately, successive governments did not meddle with the policies relating to the IT industry. The officials at the DoE provided continuity in the policy framework.

A number of concessions were given to software companies after liberalization in 1991. The import duty on computers used for software export was abolished. Software companies' export earnings were made tax free for 10 years. Multinational companies were allowed to operate in India with 100 percent equity. According to N.R. Narayana Murthy,

one of the founders of INFOSYS, a spectacularly successful IT software services company,³⁴ three other policy changes significantly altered the business environment: easier convertibility of rupees to hard currency, permission to raise capital through initial public offerings (IPOs), and the abolition of duties on imported software tools.

During this period, the government also permitted private software companies to have dedicated satellite links with their overseas customers. The National Telecom Policy 1994 (www.trai.gov.in/Content/telecom_policy_1994.aspx) allowed private companies to enter the telecommunication business. This had far-reaching consequences later, particularly for the mobile communication industry. The devaluation of the rupee enabled Indian software companies to be competitive in selling their services and MNCs to establish branches in India at low cost.

Consequences of Changes in Government Policies

With the changes in the government policies, there was a sudden spurt in the activities of Indian software companies. Earnings from exports, which were around US\$128 million during 1990–1991 went up to around US\$1.76 billion in 1997–1998,³³ an average growth of 45 percent each year. Indian software companies invented what is known as the global delivery model (GDM) and the 24-hour workday for the IT industry.³⁴ They became quality conscious and obtained certification from the International Organization for Standardization (ISO) as well as the Software Engineering Institute (SEI) at Carnegie Mellon University. The Y2K and the Euro conversion requirements created a large international market. In fact, the Y2K problem, which was considered a big threat in the mid-1990s, did not become one partly because of the large number of software professionals in India and the US working diligently to fix the bug during the period from 1993 to 1999.³⁵

The government policy of allowing foreign direct investment resulted in many multinational firms setting up software development centers in India. For example, American Express established a center in Mumbai in 1994 to carry out back-office functions such as accounts receivable, payroll processing, and inventory control. IBM, which wanted to reenter India, partnered with the Tata group and started TATA-IBM in 1992 with a 50 percent stake. Many other companies, such as HP, Oracle, and GE Capital,

began operations in India. The liberalization of communications encouraged more than 200 software services companies to set up private dedicated satellite links with their clients to develop and maintain software for them. The period 1991–1997 was one of double-digit growth.³³

Rapid Growth of the IT Industry (1998–2010)

The impact of liberalization of the IT industry and the subsequent recommendations of the IT Task Force in 2000 were felt during this period. The emphasis on hardware, computer production, and computer imports, which held center stage from 1955 to 1990, had started gradually shifting to software services during 1991–1997. The lower hardware costs, faster communications, and emergence of STPs during 1991–1997, with subsequent earning of hard currency by software services companies, caused policymakers to shift their focus to software. The passage of the Special Economic Zone (SEZ) Act in 2005 allowed duty-free import of hardware. Income tax exemption on export earnings for 10 years gave an impetus to a large number of companies to set up units for software and services export. By 2007, 257 software/service companies were set up in SEZ.⁴ The change in the rules that allowed a substantial part of export earnings by Indian IT companies to be invested abroad led to the acquisition of IT companies in western countries by Indian IT companies. Quite a few Indian IT companies became multinationals, with 10 of them listed in overseas stock exchanges. These companies had more than 400 delivery centers outside India, with a presence in 52 countries. The export earnings of Indian IT companies, which was US\$2 billion in 1998, grew to US\$50 billion in 2010.³

Political Environment and Government Policies

In the 1998 general election, a coalition called the National Democratic Alliance (NDA) with the Bharatiya Janata Party (BJP) as the majority partner, came to power. The BJP-led NDA was proactive as far as the IT industry was concerned. The export earnings of IT companies were growing around 35 percent each year. NDA lost the election in 2003. The Congress Party returned to power as a senior partner of a coalition called the United Progressive Alliance (UPA). The liberalized policies initiated in 1998 continued, and the economic growth in 2007 was 9.4 percent.³⁶

By 2010, Indian IT companies were recognized as world class based on their performance.

The UPA won the election again in 2008 and was in power until 2014.

Soon after the NDA came to power in 1998, it set up an IT task force to suggest wide-ranging reforms and incentives to the IT industry to achieve a target of export earnings of US\$50 billion by 2008. The task force gave 108 recommendations to the government (see <http://it-taskforce.nic.in>).

The other major action taken during 1998–2010 was the expansion of education in IT-related areas and engineering. Private corporate bodies were allowed to set up universities in 2002.³⁷ India's parliament passed the IT Act in 2000 to facilitate e-commerce. It was amended in 2008. India was indeed one of the first few countries to pass IT laws.

Status of the IT Industry

By 2010, Indian IT companies were recognized as world class based on their performance. From low-level testing type projects, the major companies graduated to develop end-to-end applications such as processing credit card payments. Instead of charging based on manpower cost plus expenses plus profit, companies were now taking fixed-price contracts to deliver application software of requisite quality in a specified time. Indian companies were no longer competing for software services contracts based on low cost but rather on quality and timely delivery.³⁴ The greatest advantage Indian IT companies now had was project planning experience and process maturity, as evidenced by their attaining SEI's Capability Maturity Model (CMM) level 5 certification.³⁸ In 1999, six of the 12 CMM level 5 certified companies in the world were in India. By 2010, more than 400 of the Fortune 500 companies were clients of Indian software companies.³⁹

In addition to software development companies, some new businesses that depended on software and fast worldwide communica-

tion grew rapidly during this period. These were IT-enabled services (ITeS) and business process outsourcing (BPO). ITeS included tasks such as checking insurance claims, filing income tax returns, medical transcription, remote support to fix software bugs, and manning call centers. The call centers operate 24/7 for worldwide customers and require language proficiency mostly in English and some European languages.

BPO primarily performs back-office work (such as accounts receivable, payroll processing, account reconciliation, and inventory management) for a number of organizations, the largest segment being banks and insurance companies. In 1994, American Express was the first organization to start BPO work in India; GE Capital International Services followed in 1997.³⁹ The success of these pioneers induced a large number of Indian companies to start BPO centers for foreign clients.

Another significant development in India during this period was the establishment of research, design, and development centers for several multinational companies. The centers were being set up in India to take advantage of the availability of high-quality computer science graduates at a reasonable cost. The availability of good quality office space in metropolitan areas and improved communication facilities was another incentive. The policy change of allowing companies to have 100 percent ownership without needing an Indian partner was vital (remember that IBM left India in 1978 when asked to dilute equity to 40 percent). IBM, which had returned to India in 1992 as an equal partner of the Tata group, bought off the group's share in 1999. By 2010, IBM had approximately 85,000 employees in India, second only to the number of its employees in the US. By 2010, other large multinationals operating development centers in India were Accenture, Cisco, Dell, GE, Motorola, Microsoft, Oracle, Adobe, SAP, Philips, HP, and Google. TI, which had started developing software tools in India in 1985, discovered that the quality of its employees in India was as good as that in other countries. The company started end-to-end design of ICs and followed this by introducing new products from its India center. TI's example is typical of multinationals that came to India to take advantage of the low cost. They changed their routine work to work that required design expertise when they found that the quality of engineers was good, and finally they started innovation centers. For example, by 2008, TI

had obtained 309 patents from its India center. Other microelectronics design and R&D centers were also set up in India during this period by companies such as AMD, Philips, Intel, and ST Microelectronics. The multinationals operating in India obtained more than 1,600 patents between 2006 and 2010.⁴⁰

An important impetus for the growth of IT during this period was the entry of venture capitalists and angel investors in sizable numbers. In 2007, 905 deals worth US\$5.3 billion were signed in addition to 748 Internet-specific deals worth US\$4.6 billion.⁴¹

On the hardware side, commodity PCs and laptop manufacture slowly faded. These machines were assembled from boards and other parts imported primarily from China and Taiwan and sold by several multinational companies. Indian “manufacturers” such as HCL, WIPRO, and Zenith were also assembling machines with imported kits, but their volumes were low because they could not compete with multinational companies in quality and mass manufacturing capability.

India was, however, active in designing high-performance parallel computers. In 2003, CDAC designed a parallel machine, the Param Padma, which used 248 processors and a proprietary interconnection network. Its peak speed was 992 gigaflops and it was ranked 171 in the top 500 list of high-performance computers in the world.⁴² The Tata group's Computational Research Laboratories designed a high-performance computer named Eka in 2007 with peak speed of 172 teraflops and a sustained speed of 133 Tflops. It was rated the fourth fastest computer in the world when it was announced in 2007 and was the fastest in Asia.⁴³

E-governance grew rapidly during this period as well. Citizen services such as property registration, property tax payment, and government certification used to be manual, slow, and prone to corruption. Use of computers expedited these services and reduced corruption. The income tax department is also now fully computerized.

Train ticket reservation, which was first introduced in 1986, became Web based in 2006, and by 2010 passengers could reserve their seats on any train and print their tickets at home using their Internet-connected PCs. In fact, by 2010, airlines, bus, theater, and many other tickets could be booked using the Internet (see <https://www.irctc.co.in>).

All these achievements would not have been possible without educated workers. There was a rapid expansion of engineering

colleges between 1998 and 2010. The groundwork for the expansion of IT education was laid in 1980 with the introduction of MCA courses and the expansion of computer science undergraduate courses. All major IT companies had in-house training programs ranging from 12 weeks to six months. The training schedule was gigantic as every major software company was recruiting around 8,000 graduates each year.

Conclusions

What did India learn from history during the period 1955–2010? Can the growth of the IT industry between 1991 and 2010 be sustained?

As a poor country with a low demand for computers, it was strategically incorrect during 1970–1980 to try to design computers starting at the component level and hope to be completely self-reliant. It would have been wiser to spend the available scarce resources in systems engineering and to build computers using subassemblies during the early stages of development. This is particularly true in the area of computers, where wealthier countries were making huge investments in R&D, with consequent fast changes in technology and rapid obsolescence.⁴⁴ This was realized only in the mid-1980s and the consequent policy change led to a rapid growth of computer availability. The fear of unemployment that was in the background while making decisions regarding the use of computers in the 1970s was misconstrued. In a country with a huge population and voluminous data-processing requirements, one cannot manage without computers. This was amply demonstrated when the Indian Railways computerized reservations in 1986. This project changed the mindset of both the general public and the politicians on computers. Another policy error was not to invest sufficiently in the manufacture of ICs in the late 1970s and seriously attempt to get technical collaboration from leading manufacturers in this area.

The investment made in higher education paid off handsomely. Graduates of IITs during the 1960s became technology leaders and entrepreneurs not only in India but also in the US in the 1980s. The DoE along with the Ministry of Human Resources Development took the initiative in the early 1980s to increase the human resources availability by starting bachelor's degree courses in computer science and MCA courses. A training program for teachers of computer science was

also started in the 1980s. Private companies provided vocational training. The DoE initiated an accreditation program and administered standardized examinations to certify the vocational courses given by private institutions. These early initiatives provided the human resources that enabled the software industry to take off in the 1990s.

Research projects funded by the Government of India and the United Nations Development Program, which included the National Informatics Center, the National Center for Software Technology, the Center for Development of Advanced Computing, the Computer Aided Design Centers, the Centers for Knowledge Based Computer Systems Development, and the Education and Research in Computer Networking, among others, created a large pool of technology leaders and strengthened institutional infrastructure. Compared with the investment in education and research, the investment made in computer manufacture by the government companies did not have the same multiplier effect. The investment, however, did meet some strategic requirements in defense and atomic energy.

A slew of liberalized policy initiatives taken in the mid-1980s and the early 1990s led to an exponential growth of IT companies in India. The emphasis on quality certification, systematizing application software development processes, and project management were all essential ingredients for the success of the Indian software services companies in the international market.

IT diffusion in India has been primarily in the urban areas among the middle class population. Most public services offered by the state and central government are now computerized. Banking is also computerized. Diffusion of IT to rural areas, where more than 70 percent of Indians live, has been spotty because of the poor availability of electricity, high cost of computers, and lack of availability of local language-based applications. On the other hand, almost everyone in rural areas uses mobile phones because the cost and usage charges are among the lowest in the world,⁴⁵ they do not require continuous availability of electricity, and their perceived value is high.

To sustain growth, the Indian software industry should invest more in R&D and develop innovative products not only for international markets but also for the huge local market, which has remained unexplored. The industry should also explore

innovative applications based on mobile phones for the local market.

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